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QUANTITATIVE FINANCE

The Price of Being in a Hurry

A large order cannot be traded without moving the price against itself, and the whole theory of how to do it gracefully rests on one exponent that turns out to be wrong.

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Every institutional trading desk keeps two sets of numbers. The first is the return of the portfolio the manager decided to hold. The second is the return of the portfolio the desk actually built. André Perold gave the gap between them a name in 1988 — the *implementation shortfall* — and the name stuck because the gap turned out to be embarrassingly large and unnervingly systematic [1]. It was not commissions. It was not bad luck. It was the plain fact that a decision to buy a great many shares is a decision to make those shares more expensive.

This is the awkward part of finance that the elegant parts of finance quietly assume away. Option pricing takes the price process as given. Portfolio theory takes returns and covariances as given. Both are theories of a price-taker in a market deep enough not to notice. But a pension fund rebalancing a hundred-million-dollar position is not a price-taker, and the question of how to get into a position without ruining the reason for holding it turns out to have a mathematically beautiful answer, an unusually clean piece of engineering built on top of it, and a foundation that thirty years of data have steadily undermined.

Where the Straight Line Comes From

The modern account of why trading moves prices begins with information rather than with physical scarcity. In Lawrence Glosten and Paul Milgrom's 1985 model, a market maker who must quote a price to anyone who asks faces a hazard: some of the people who ask know something [3]. A buy order is weak evidence that the asset is worth more than the current quote. The maker cannot tell an informed buyer from a liquidity-driven one, so the only defence is to quote a spread wide enough that the losses to the informed are paid for by the gains from everyone else. The spread appears even when the market maker is risk-neutral and earns nothing on average.

Albert Kyle, in the same year, turned that intuition into a quantity [2]. His setup is austere enough to fit in a paragraph. An asset is worth v, drawn from a normal distribution with mean p_0 and variance Σ_0. A single informed trader learns v exactly and chooses a quantity x. Uninformed order flow u arrives independently, normal with variance σ_u^2. A competitive market maker sees only the total $y = x + u$ and must set a price. Kyle looked for an equilibrium in which the maker's rule is linear, $p = p_0 + \lambda y$, and the informed trader's demand is linear too, $x = \beta(v - p_0)$.

Each side's problem is then one line. The informed trader maximises expected profit $x(v - p_0 - \lambda x)$, which gives $x = (v - p_0)/2\lambda$, so $\beta = 1/2\lambda$. The market maker, being competitive, must set $p = \mathbb{E}[v \mid y]$, which for jointly normal variables is a regression: $\lambda = \beta\Sigma_0/(\beta^2\Sigma_0 + \sigma_u^2)$. Substituting one into the other and solving gives

$$\lambda = \frac{1}{2}\frac{\sigma_0}{\sigma_u}, \qquad \beta = \frac{\sigma_u}{\sigma_0}, \qquad \lambda\beta = \frac{1}{2},$$

where $\sigma_0 = \sqrt{\Sigma_0}$. Two consequences follow immediately and are worth more than the algebra. The residual uncertainty after trading is $\Sigma_1 = \Sigma_0/2$: exactly half of the insider's private information ends up in the price, no matter how good the information is or how much noise there is to hide behind. And the insider's expected profit is $\sigma_0\sigma_u/2$ — it scales with the volume of noise trading, because noise is the camouflage that makes private information worth having. In the continuous-time version of the same model, λ is constant through the day and *all* of the private information is in the price by the close [2].

The number λ has had a long career since. It is the slope of price against signed order flow, it is measurable, and its reciprocal is the market's depth in shares per dollar of price movement. Joel Hasbrouck's 1991 vector-autoregression estimates of the price impact of trade innovations are the canonical empirical implementation [4]. For a 50-dollar stock with five million shares of average daily volume and 2 per cent daily volatility, plausible inputs give $\lambda \approx 2 \times 10^{-7}$ dollars per share, or a depth of about five million shares per dollar: a million shares of net imbalance moves the price twenty cents.

The crucial structural feature is that impact is *linear* in quantity. Kyle did not assume this; it fell out of normality. And because it fell out of something so innocuous, it was easy to believe.

The Schedule That Follows From a Straight Line

Grant linear impact and the execution problem becomes tractable. Dimitris Bertsimas and Andrew Lo showed in 1998 that a risk-neutral trader facing linear permanent impact should split an order into equal pieces and trade them at an even rate [5]. This is the origin of the ubiquitous time-weighted average price algorithm, and it is also a warning, because it says the trader should be indifferent to taking all day.

Real traders are not indifferent, because waiting is exposure. Robert Almgren and Neil Chriss added precisely that in 2001 [6]. Let $x(t)$ be the shares still held while liquidating X shares over $[0, T]$, so the trading rate is $-\dot{x}$. Permanent impact γ per share shifts the price for good; temporary impact η per unit rate is the concession paid to trade quickly. Then the expected cost is $\gamma X^2/2 + \eta \int_0^T \dot{x}^2\,dt$ and its variance is $\sigma^2 \int_0^T x^2\,dt$, and minimising expectation plus λ times variance is a textbook calculus-of-variations problem whose Euler-Lagrange equation is $\ddot{x} = \kappa^2 x$ with

$$\kappa = \sqrt{\frac{\lambda\sigma^2}{\tilde{\eta}}}, \qquad x(t) = X\,\frac{\sinh\big(\kappa(T-t)\big)}{\sinh(\kappa T)}.$$

The tilde on η is Almgren and Chriss's small discrete-time correction, $\tilde{\eta} = \eta - \gamma\tau/2$ for intervals of length τ. Solving the discretised problem exactly on a four-thousand-step grid reproduces the closed form to nine parts in a billion, so the hyperbolic sine is not an approximation to the answer; it is the answer.

The engineering payoff is real. On a one-million-share order in the stock above, the optimal path accepts about 12.5 basis points more expected cost than an even schedule in exchange for a 32 per cent reduction in the standard deviation of the outcome. Almgren and Chriss's contribution was to draw the whole efficient frontier of that trade-off, which is why the paper is on every execution desk's shelf.

But the striking feature of the formula is the thing that is *not* in it.

> **WHAT DOES THE OPTIMAL EXECUTION SPEED DEPEND ON?**
>
> The rate constant κ contains risk aversion, volatility and the impact coefficient. It does not contain X. So the half-life of the position, $\ln 2/\kappa$ in the long-horizon limit, is identical for an order of fifty thousand shares and an order of five million. Almgren and Chriss note this explicitly: $\theta = 1/\kappa$ is an intrinsic time scale of the trader and the stock, not of the trade. Under the calibration used throughout here — 2 per cent daily volatility, five million shares of daily volume, risk aversion such that $\kappa = \sqrt{5}$ per day — that half-life is 2.015 trading hours, flat across three decades of order size. Re-solving the discrete problem at three sizes gives normalised trajectories that agree to eight decimal places.
>
> That is an extraordinary claim about the world, and it is the model's, not a caricature of it.

What the Data Actually Say

The claim is also false, and the reason is that impact is not linear.

The first systematic evidence came from an unexpected direction. Fabrizio Lillo, Doyne Farmer and Rosario Mantegna, writing in *Nature* in 2003, rescaled the average price change per trade by a market-capitalisation-dependent liquidity factor and found that the thousand largest stocks on the New York Stock Exchange collapsed onto a single master curve for four separate years [9]. The curve rises roughly as the square root of volume for small trades and flattens further for large ones. It is concave, not straight.

Two years later Almgren, Thum, Hauptmann and Li took nearly 700,000 orders from Citigroup's US equity desks over the nineteen months from December 2001 to June 2003 and fitted the temporary impact directly as a function of trade rate. They rejected the square root in favour of a 3/5 power law [8]. Note the shape of that sentence: the debate by 2005 was no longer whether impact was linear, but which concave exponent fitted best.

The most durable version of the finding concerns *metaorders* — a manager's whole intended position, executed over hours or days by an algorithm through hundreds of child orders. Across equities, futures, options and even Bitcoin, the average impact of a metaorder of Q shares in an instrument with daily volume V and daily volatility σ is

$$I(Q) \simeq Y\sigma\sqrt{Q/V},$$

with Y a dimensionless constant of order one — estimates cluster near 0.5 for futures and nearer 0.9 for equities, and a 2026 study of 178 days of Nasdaq order-by-order data for a single large-cap stock recovers a raw prefactor of 0.69 from anonymous order flow alone [11], [18]. Bence Tóth and co-authors argued in 2011 that this is not a curiosity but a symptom: the visible order book holds a negligible fraction of the day's volume, the true liquidity is *latent* and vanishes linearly around the current price, and a market in that state sits at a critical point where linear response breaks down [11]. Emiliano Zarinelli and colleagues later found a logarithmic form fits US metaorders from 2007 to 2009 better still [12] — concavity is not in doubt, only its exact functional form.

Concavity has consequences in both directions, and they pull opposite ways. Below is the same instrument under a linear model calibrated to Kyle and under the square-root law with $Y = 0.5$, both at 2 per cent daily volatility.

Order, per cent of volume	Linear, bp	Square root, bp	Ratio
0.1	0.20	3.16	15.8
1	2.00	10.00	5.0
5	10.00	22.36	2.2
25	50.00	50.00	1.0
100	200.00	100.00	0.5

The two agree exactly where $Q/V = Y^2$, here 25 per cent of a day's volume. Below that, concavity makes trading *dearer* than linearity predicts; above it, cheaper. In the limit the square-root law is stranger still: the marginal impact of the first share, $dI/dQ \propto Q^{-1/2}$, diverges. There is no such thing as a trade small enough to be free. Against this, the largest sample of real executions ever published — 1.7 trillion dollars of live trades from a single institutional manager across 21 developed markets over nineteen years — reports mean market impact near 10 basis points and mean implementation shortfall near 11, with medians several basis points lower, which is an order of magnitude below what much of the academic literature had assumed [17].

The Exponent Is Not a Detail

It would be comfortable to treat the exponent as a refinement — the same schedule, slightly re-tuned. Almgren himself extended the framework to nonlinear impact functions and trading-enhanced risk as early as 2003, and the machinery survives [7]. But the exponent turns out to control whether the schedule depends on order size at all, and that can be shown in closed form.

Suppose the concession for trading at rate w is $\eta_k w^k$, so the cost rate is $\eta_k w^{1+k}$, and keep the Almgren-Chriss risk penalty. Liquidating with no deadline means minimising

$$J \;=\; \int_0^{\infty} \Big[\eta_k\, |\dot{x}|^{1+k} + \lambda\sigma^2 x^2\Big]\, dt.$$

The integrand carries no explicit time, so the associated Hamiltonian is conserved, and the boundary condition at infinity forces it to vanish. With $w = -\dot{x}$ that gives a first integral of remarkable simplicity,

$$\eta_k\, k\, w^{1+k} = \lambda\sigma^2 x^2 \quad\Longrightarrow\quad w = c\, x^{2/(1+k)},$$

with $c = (\lambda\sigma^2/\eta_k k)^{1/(1+k)}$. At $k = 1$ this is $w = \kappa x$: exponential decay, the long-horizon limit of the hyperbolic sine, at a rate free of X. For any other k it separates, and writing $p = 2/(1+k)$ the half-life is

$$\theta \;=\; \frac{X^{\,1-p}\big(2^{\,p-1} - 1\big)}{(p-1)\, c} \;\;\propto\;\; X^{(k-1)/(k+1)}.$$

So the size-independence of the Almgren-Chriss horizon is not a discovery about markets. It is the single value $k = 1$ at which the exponent $(k-1)/(k+1)$ happens to vanish. At the measured $k = 3/5$ the exponent is exactly $-1/4$; at $k = 1/2$ it is exactly $-1/3$; for convex impact, $k > 1$, it turns positive and larger orders should be worked more slowly. Integrating the rate equation numerically agrees with the closed form to a part in a million, and letting $k \to 1$ recovers $\ln 2/\kappa$ continuously.

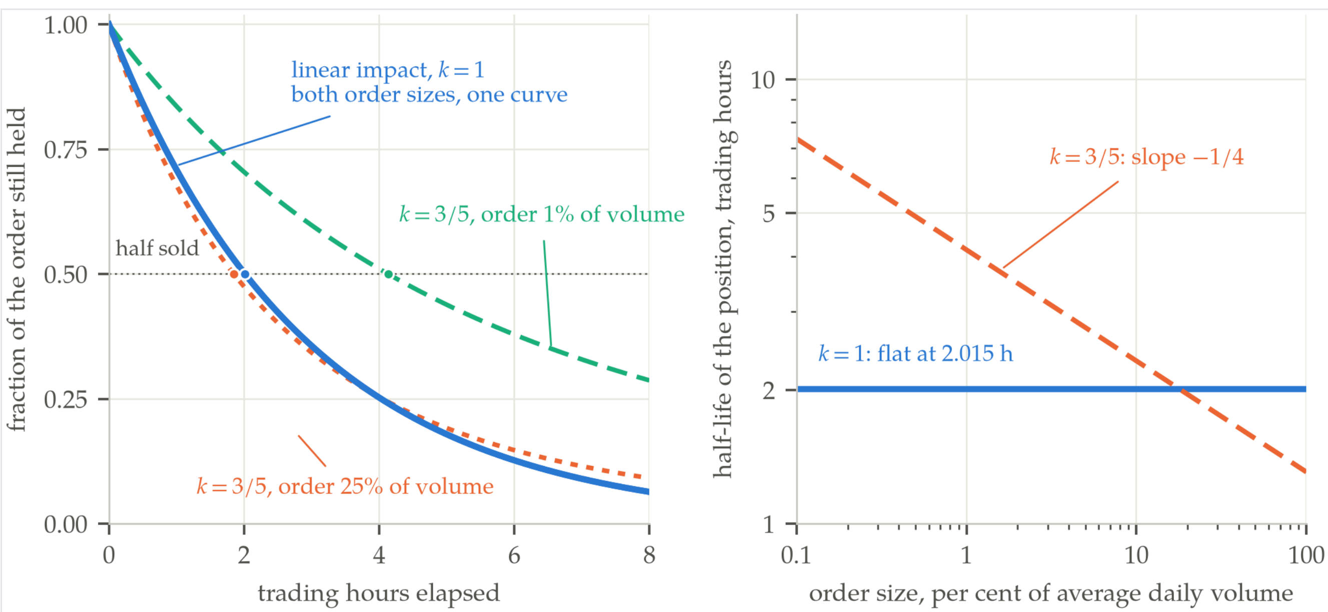

FIGURE Under linear impact the optimal liquidation path is the same shape whatever the order size, and its half-life is a constant 2.015 trading hours. Under the measured exponent the collapse fails: a 25-times-larger order is worked 2.24 times faster, and the half-life falls as the fourth root of size. Calibration in figures/017-execution-trajectories.py.

The sign surprises people, so it is worth saying why it comes out that way. Concave impact means doubling the trading rate less than doubles the concession per share, so speed is relatively cheap. Meanwhile the risk of still holding the position grows with the square of what is left. For a large enough position the risk term wins, and the optimal policy is to trade a *larger fraction* of the remaining stock per unit time: $w/x \propto x^{(1-k)/(1+k)}$, which is constant only at $k = 1$.

Order, per cent of volume	Half-life, $k = 1$	Half-life, $k = 3/5$
1	2.015 h	4.134 h
5	2.015 h	2.765 h
25	2.015 h	1.849 h

Two caveats belong with that table. It holds risk aversion fixed while size varies, which is a modelling convention rather than a fact about traders. And at 25 per cent of daily volume the model's own initial participation rate exceeds half the day's volume, which is beyond where a single-agent impact model deserves to be believed. The scaling exponent is exact; the hours are an illustration.

The Constraint That Cannot Be Dropped

If impact is concave, why not simply write concave impact into the model everywhere and be done? Because a piece of the structure is not free to move.

Gur Huberman and Werner Stanzl proved in 2004 that if price impact is permanent and time-independent, only a *linear* impact function rules out quasi-arbitrage — a sequence of round trips generating unbounded expected profit at unbounded Sharpe ratio [13]. Any curvature in the permanent component can be pumped. Jim Gatheral sharpened the point in 2010 from a no-dynamic-arbitrage principle requiring that round-trip costs be non-negative on average, and derived a relationship between the shape of the impact function and the shape of its decay: the widely assumed exponential decay of impact is compatible only with linear impact, while a power-law impact function requires power-law decay [14]. The empirical inequalities, he noted, are close to equalities.

The resolution is that impact is not one thing. There is a permanent component, which theory says must be linear and which is the informational residue Kyle described, and a transient component that decays, in which all the measured concavity lives. Anna Obizhaeva and Jiang Wang built the canonical model of the transient side in 2013, replacing the static impact coefficient with a limit order book that is depleted by trading and refills at a finite rate. Their optimal strategy is not smooth at all: a discrete block at the start to push the book away from its steady state and attract fresh liquidity, a continuous middle phase picking off the inflow, and a final discrete block when future supply no longer matters [15]. The propagator models of Jean-Philippe Bouchaud and co-authors describe the same decay empirically, and find a market sitting exactly at the boundary where prices are diffusive and the average response is nearly flat [10].

There is even a route back to the square root from pure scaling. Kyle and Obizhaeva's market microstructure invariance proposes that the distribution of bets and their costs is constant across assets when measured per unit of *business* time rather than clock time; combined with dimensional analysis and leverage neutrality, the square-root law emerges as a special case, and the resulting predictions survive testing on more than 400,000 portfolio transition orders [16].

That is a satisfying place for the field to have arrived, and an uncomfortable one. The most-used execution model in the industry is exactly solvable because of an assumption that the same industry's data rejected within four years of its publication, and its most quoted qualitative lesson — that how fast you should trade has nothing to do with how much you have to trade — is an artefact of that assumption sitting at the one exponent where the dependence cancels. The model remains useful, which is the ordinary fate of a good approximation. But it is worth knowing which of its conclusions are about markets and which are about the number one.

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